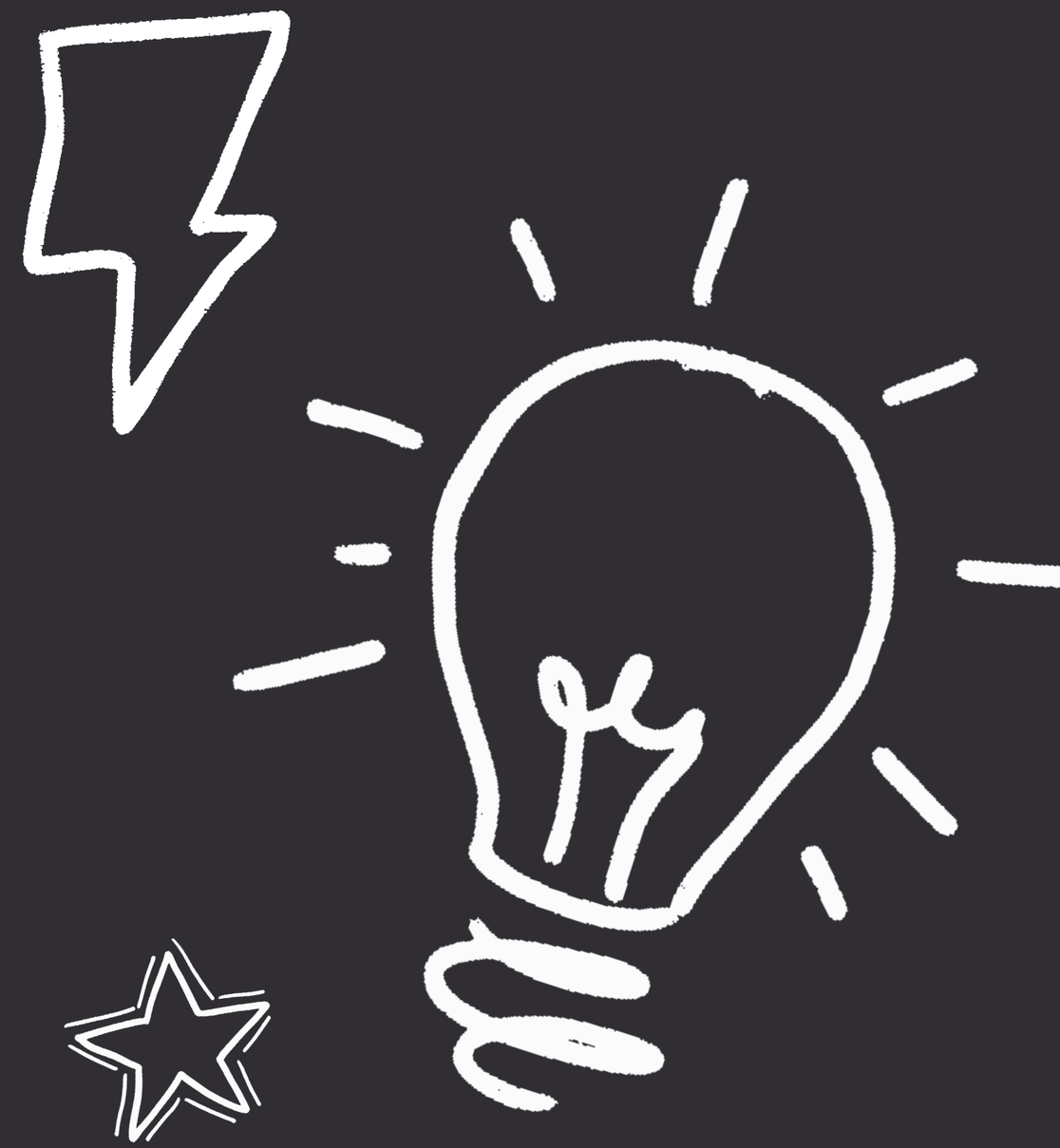
# Programarea functionala



# Interfata functionala

Java defineste “interfata functionala” o interfata cu o singura metoda abstracta. In alte cuvinte, o interfata cu o singura functionalitate.



Main.java

```
class Tiger extends Animal {

    @Override
    public void sprint() {
        System.out.println("Tiger runs fast");
    }
}
class Turtle extends Animal implements Sprint {

    @Override
    public void sprint() {
        System.out.println("Turtle runs slow");
    }
}
public class Main {
    public static void main(String[] args) {
        Animal tiger = new Tiger();
        Animal turtle = new Turtle();

        tiger.sprint();
        turtle.sprint();

    }
}
```

Animal.java

```
1  package com.company;
2
3  public abstract class Animal implements Sprint{
4  }
5
```

Sprint.java

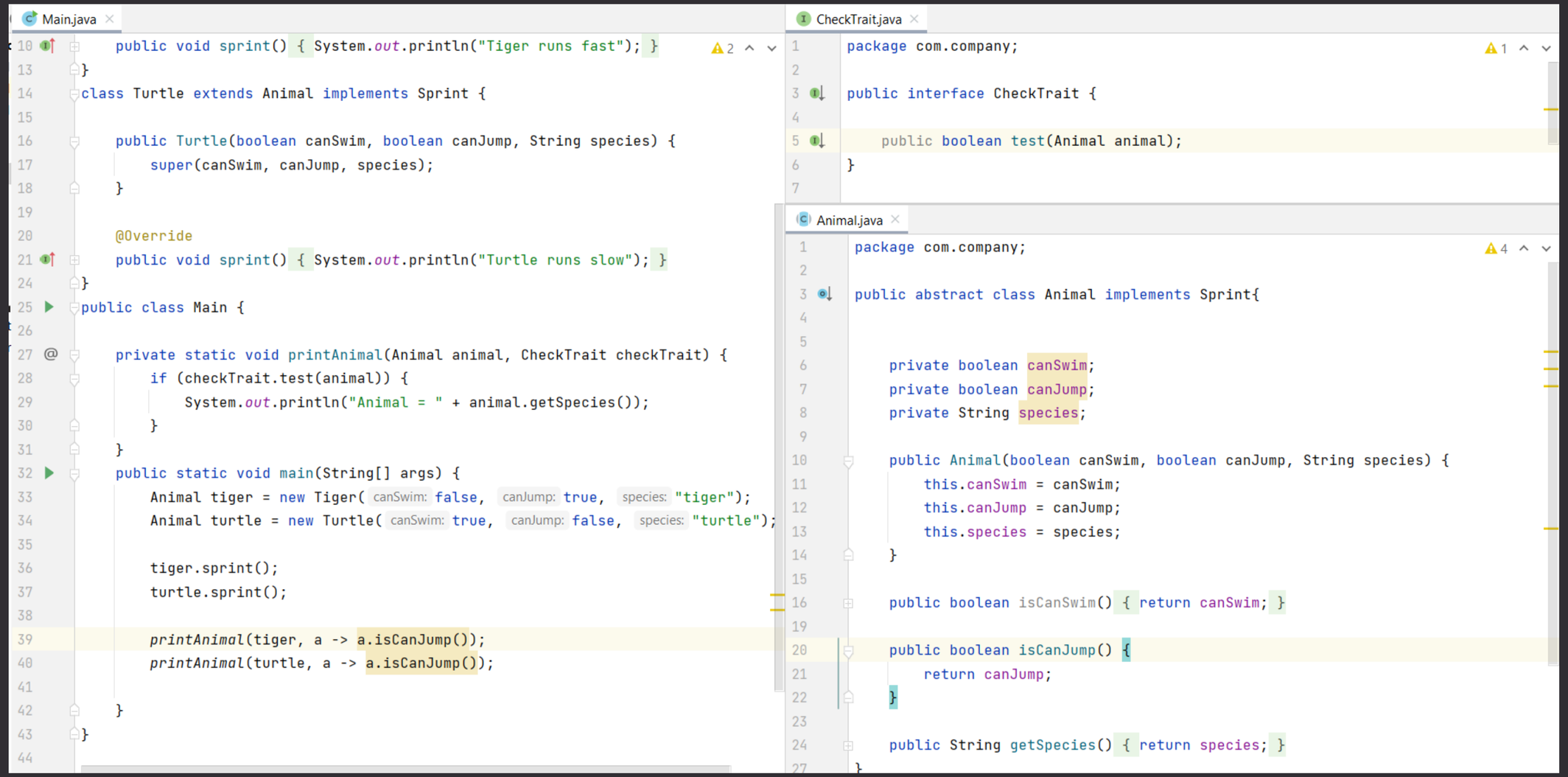
```
1  package com.company;
2
3  public interface Sprint {
4      void sprint() ;
5  }
```

# Expresii lambda

## Lambda Syntax

- No arguments:
```
() -> System.out.println("Hello")
```
- One argument:
```
s -> System.out.println(s)
```
- Two arguments:
```
(x, y) -> x + y
```
- With explicit argument types:
```
(Integer x, Integer y) -> x + y
```
- Multiple statements:
```
(x, y) -> {
    System.out.println(x);
    System.out.println(y);
    return (x + y);
}
```

# Implementarea interfetelor functionale folosind expresii lambda



Main.java

```
    public void sprint() { System.out.println("Tiger runs fast"); }
}
class Turtle extends Animal implements Sprint {

    public Turtle(boolean canSwim, boolean canJump, String species) {
        super(canSwim, canJump, species);
    }

    @Override
    public void sprint() { System.out.println("Turtle runs slow"); }
}
public class Main {

    private static void printAnimal(Animal animal, CheckTrait checkTrait) {
        if (checkTrait.test(animal)) {
            System.out.println("Animal = " + animal.getSpecies());
        }
    }
    public static void main(String[] args) {
        Animal tiger = new Tiger( canSwim: false,  canJump: true,  species: "tiger");
        Animal turtle = new Turtle( canSwim: true,  canJump: false,  species: "turtle");

        tiger.sprint();
        turtle.sprint();

        printAnimal(tiger, a -> a.isCanJump());
        printAnimal(turtle, a -> a.isCanJump());

    }
}
```

CheckTrait.java

```
package com.company;

public interface CheckTrait {

    public boolean test(Animal animal);
}
```

Animal.java

```
package com.company;

public abstract class Animal implements Sprint{


    private boolean canSwim;
    private boolean canJump;
    private String species;

    public Animal(boolean canSwim, boolean canJump, String species) {
        this.canSwim = canSwim;
        this.canJump = canJump;
        this.species = species;
    }

    public boolean isCanSwim() { return canSwim; }

    public boolean isCanJump() {
        return canJump;
    }

    public String getSpecies() { return species; }
}
```

# Interfata Predicate<T>

```
    private static void printAnimalPredicate(Animal animal, Predicate<Animal> checkTrait) {
        if (checkTrait.test(animal)) {
            System.out.println("Animal = " + animal.getSpecies());
        }
    }
    public static void main(String[] args) {
        Animal tiger = new Tiger( canSwim: false,  canJump: true,  species: "tiger");
        Animal turtle = new Turtle( canSwim: true,  canJump: false,  species: "turtle");

        tiger.sprint();
        turtle.sprint();

//          printAnimal(tiger, a -> a.isCanJump());
//          printAnimal(turtle, a -> a.isCanJump());

        printAnimalPredicate(tiger, a -> a.isCanJump());
        printAnimalPredicate(turtle, a -> a.isCanJump());

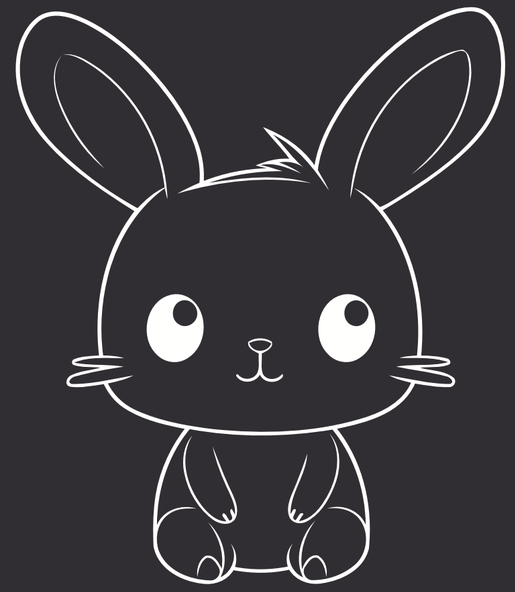
    }
```

# Now some work...



# Ferma animalelor

- Adaptati scenariul anterior despre ferma animalelor astfel incat sa puneti in practica notiunile nou invatate.
- Ex: Pentru hranirea animalelor se poate crea o interfata Hranire cu o singura metoda abstracta hraneste() care va fi responsabila de hranirea particulara a fiecarui animal.



# Resurse

- https://www.baeldung.com/java-8-lambda-expressions-tips
- https://www.geeksforgeeks.org/java-8-predicate-with-examples/

